(notes) 6.6 Equations with Logs and Exponents, part 2

$$\log_{base} n = exp \longleftrightarrow base^{exp} = n$$

1. x is in the exponent

ex. $3^x = 91$

1. change to log form
2. If the base is not 10 or e, use change of base

   (change the base to be 10 or e!)
3. Put it in calculator

2. x is the base

Won't really see this in this section but....

1. Change to exponent
2. Undo the exponent with the exponential reciprocal of the exponent

3. x is the number/result

ex. $\log_2(3x) = 7$

1. Change to exponent form
2. Calculate the base to the exponent to get a number
3. Solve for x

x is divided, it occurs in two positions

1. If the logs are added, use product rule to combine logs
2. If the logs are subtracted, use the quotient rule to combine logs
3. Convert to exponent form
4. Solve for x

ex. $\log(x+4) - \log(x+2) = 2$

$$\log(x+4) - \log(x+2) = 2 \qquad \rightarrow \text{combine logs w same base}$$

$$\log\left(\frac{x+4}{x+2}\right) = 2 \qquad \rightarrow \text{used quotient rule}$$

now we have x in number/result position!

change to exponent form!

recall $\log$ w no stated base is the common log w base 10

$$10^2 = \left(\frac{x+4}{x+2}\right) \qquad \rightarrow \text{exp form, } \log() \text{ has base 10}$$

now solve for x.....

$$100 = \frac{x+4}{x+2}$$

$$100(x+2) = x+4 \qquad \rightarrow \text{solve for } x$$

$$100x + 200 = x + 4 \qquad \rightarrow \text{distribute 100}$$

$$100x - \boxed{x} + 200 = x - \boxed{x} + 4$$

$$99x + 200 = 4$$

$$99x + 200 - \boxed{200} = 4 - \boxed{200}$$

$$99x = -196$$

$$\frac{99x}{99} = -\frac{196}{99}$$

$$x = -\frac{196}{99}$$

ex. $\log_{45} x + \log_{45}(x+4) = 1$ **(tricky)**

$$\log_{45} x + \log(x+4) = 1 \qquad \rightarrow \text{combine logs w same base}$$

$$\log_{45}(x \cdot (x+4)) = 1 \qquad \rightarrow \text{use product rule}$$

x is in result position, change to exponent form!

$$45^1 = x \cdot (x+4) \qquad \rightarrow \text{switch to exp form: base=45, exp=1}$$

now solve for x.....

$$45 = x^2 + 4x$$

$$0 = x^2 + 4x - 45$$

to solve for x, we factor the quadratic!

$$0 = (x+9)(x-5)$$

using zero product property:

$$x + 9 = 0 \quad x - 5 = 0$$

$$\implies x = -9 \quad x = 5$$

note: domain for this equation is $(0, \infty)$

so only $x = 5$ is the answer...

This problem is tricky because one of our answers is not in the domain. To determine the domain, we look at the two logs. From their individual graphs, we know there will be domain restrictions. The first log ($\log_{45} x$) has the domain of $(0, \infty)$ and second log ($\log_{45}(x+4)$) has the domain of $(-4, \infty)$. So the overall domain is the more restricted domain of the two logs which will be $(0, \infty)$. Since we got $x = -9, 5$, the negative answer is not in the domain and must be discarded 😢

Practice, solve for x

1. $\log(x+5) - \log(x+1) = 1$
2. $\log_{42} x + \log_{42}(x-1) = 1$ 😟

Word Problems

ex. The pH scale for acidity is defined by $\mathrm{pH} = -\log[H^+]$ where $[H^+]$ is the concentration of hydrogen ions measured in moles per liter (M).
A solution has a pH of 11.25.
Calculate the concentration of hydrogen ions in moles per liter (M).

1. Start with the equation and the list of knowns!

$\mathrm{pH} = \log[H^+]$

$\mathrm{pH} = 11.25$ ($4000 is deposited)

$[H^+] =$????? (unknown to solve for!)

2. Plug in values into pH formula

$$\begin{aligned}
\mathrm{pH} &= -\log[H^+] && \rightarrow \text{pH formula} \\
11.25 &= -\log x && \rightarrow \text{plug in values, } x \text{ unknown} \\
-11.25 &= \log x && \rightarrow \text{isolate the } \log() \\
&\text{unknown is in result position...} \\
&\text{convert to exponent form!} \\
10^{-11.25} &= x && \rightarrow \text{common } \log() \text{ has base 10, exponent: } -11.25 \\
x &= 5.623 \times 10^{-12} && \rightarrow \text{get this with calculator} \\
&\implies \text{concetration is } 5.623 \times 10^{-12}\ \mathrm{M}
\end{aligned}$$

ex. A population of bacteria is growing according to the equation $P(t) = 600e^{0.11t}$. Estimate when the population will exceed 1213.

1. Identify the information given and the value you need to find.

   $P = 600$

   $e = 2.71828\ldots$ (constant value, use e button on calculator)

   $r = 0.11$

   $t =$???? (the unknown to solve for!)

   $P(t) = 1213$

2. Plug in values into continuous interest/growth formula

$$
\begin{aligned}
P(t) &= Pe^{rt} \qquad \rightarrow \text{continuous growth/interest formula} \\
&\text{plug in values, } t \text{ unknown} \\
1213 &= 600e^{0.11*t} \\
&\text{isolate the } e^{0.11*t} \\
&\text{divide both sides by } 600 \\
\frac{1213}{600} &= \frac{600e^{0.11*t}}{600} \qquad \rightarrow \text{isolate } t \text{ in exp} \\
2.02 &= e^{0.11*t} \\
&t \text{ in exponent so change to log form!} \\
&\text{change to log form :)} \\
&\text{since base is } e\text{, use } \ln(), \text{ exponent is } (0.11 * t) \\
\ln 2.02 &= (0.11 * t) \qquad \rightarrow \text{log form} \\
\frac{\ln 2.02}{0.11} &= \frac{0.11 * t}{0.11} \\
t &= \frac{\ln 2}{0.03} \qquad \rightarrow \text{put in calculator} \\
t &= 6.400 \implies 6.4 \text{ years for bacteria to exceed a pop of } 1213
\end{aligned}
$$

Practice word problems

1. The pH scale for acidity is defined by $\text{pH} = -\log[H^+]$ where $[H^+]$ is the concentration of hydrogen ions measured in moles per liter (M).
   A solution has a pH of 1.45.
   Calculate the concentration of hydrogen ions in moles per liter (M).
   (answer: 0.03548133 M)

2. $8000 is deposited in an account earning 4% interest compounded continuously. Use the continuous interest formula $A = Pe^{rt}$ to determine how long it takes for the amount in the account to double. Round answer to 2 decimal places.
   (answer: 17.33 years)